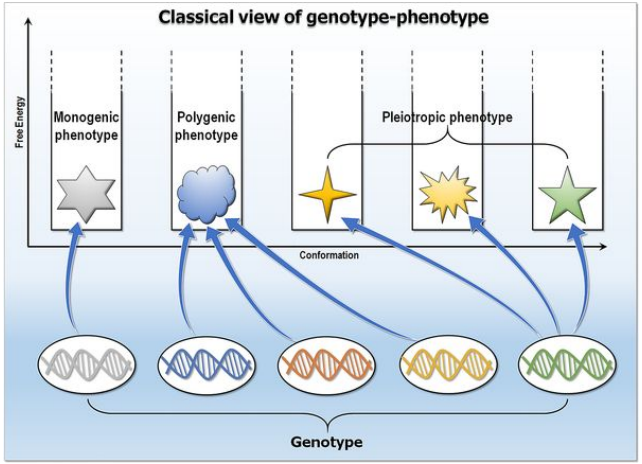
BME 105: Genetics in the Genomics Era

Lecture 13: Linking genotype to phenotype



Instructor: Dr. Shelbi Russell,
Assistant Professor

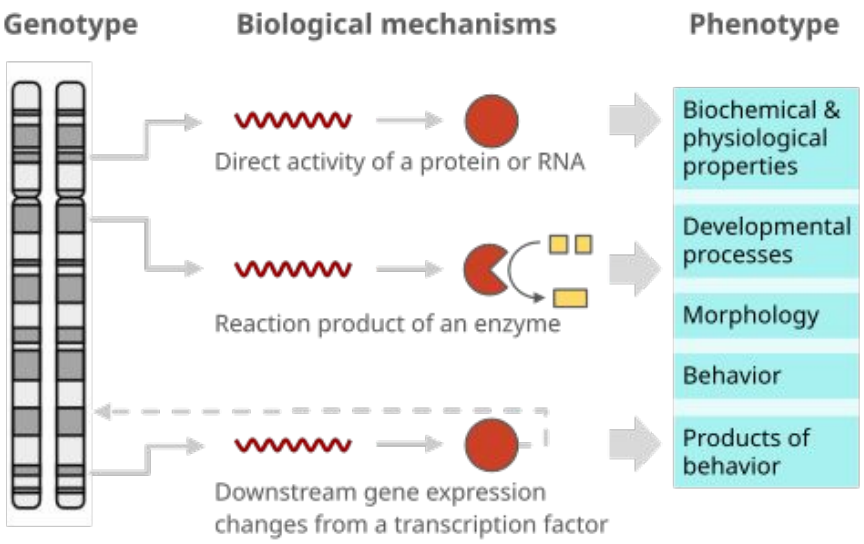
Course website on canvas.ucsc.edu

Lecture 13 Learning Objectives

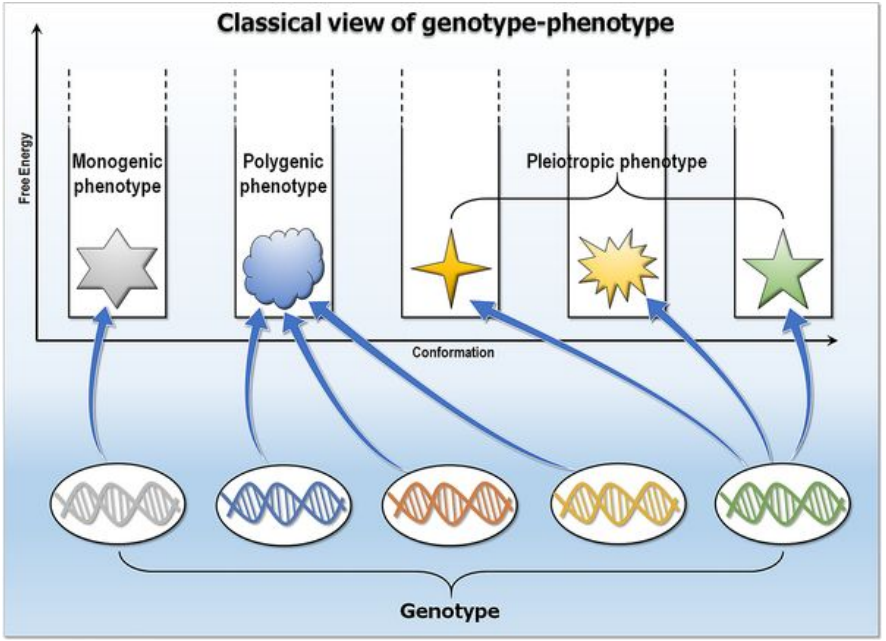
- Understand the two ways genotype is linked to phenotype
  - Biology links genotype to phenotype
  - Research works to discover how biology works
- Begin to understand how researchers discover genetic mechanisms
  - Forward genetics: phenotype motivates discovery of underpinning genotype
  - Reverse genetics: genotype motivates testing effects on phenotype
- Be able to use complementation tests to determine the number of <u>unlinked</u> genes contributing to a phenotype
  - Understand why recombination is the limiting factor to fine-scale trait mapping genome-wide

Two interpretations of “linking genotype to phenotype”

How <u>biology</u> links geno-pheno:

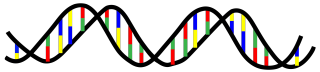


How <u>researchers</u> link geno-pheno:

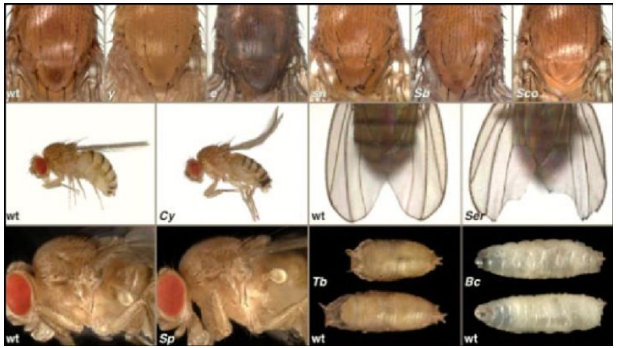


Forward genetics: start with phenotype, discover genotype

Historically, we started with these approaches



Genotype(s) → phenotype(s)



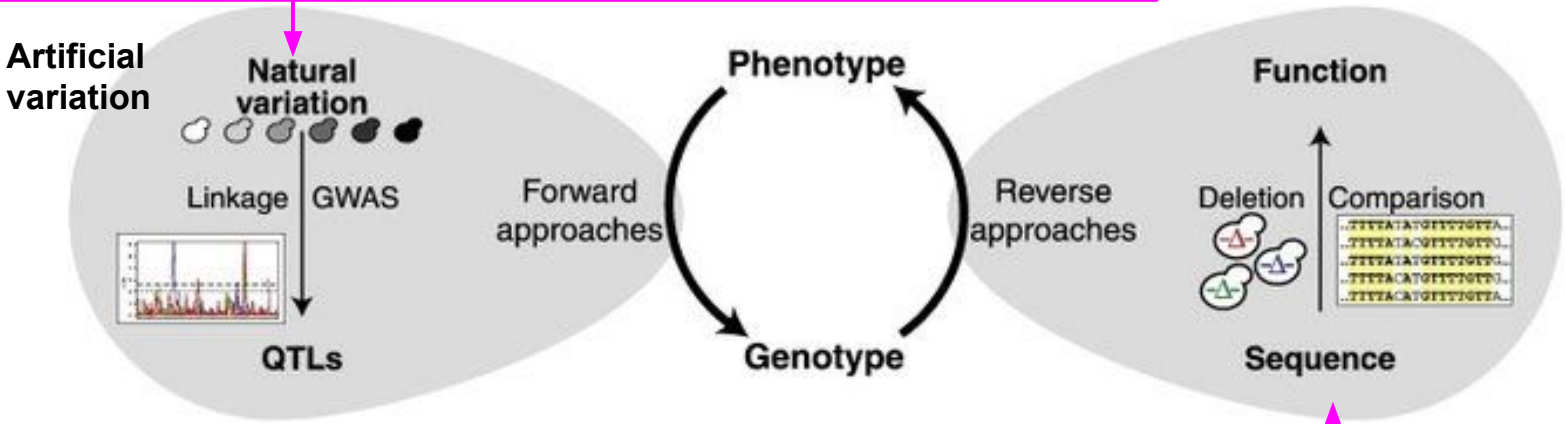
These approaches have expanded with multi-omics datasets

Reverse genetics: start with genotype, discover phenotype

Linking genotype to phenotype with -omics approaches

Forward samples:

- Collected from natural populations
- Sampled from nature, inbred in lab, recombined to unlink alleles
- Random mutagenesis
- Positional cloning

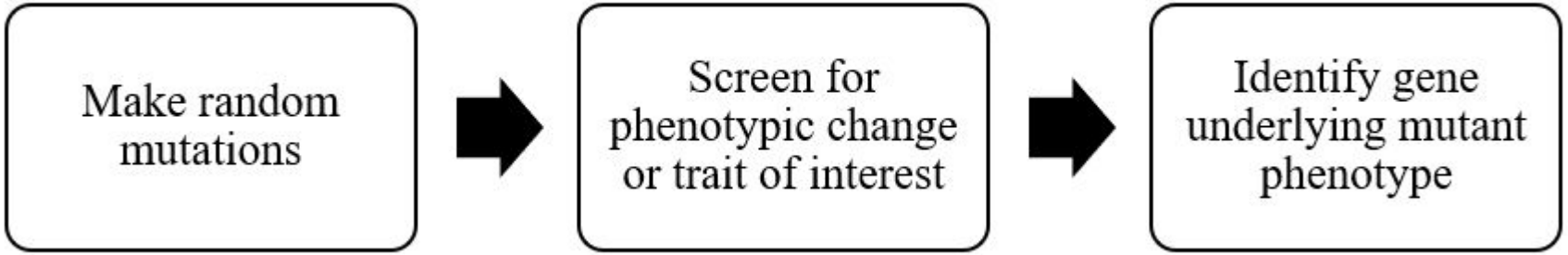


Reverse samples:

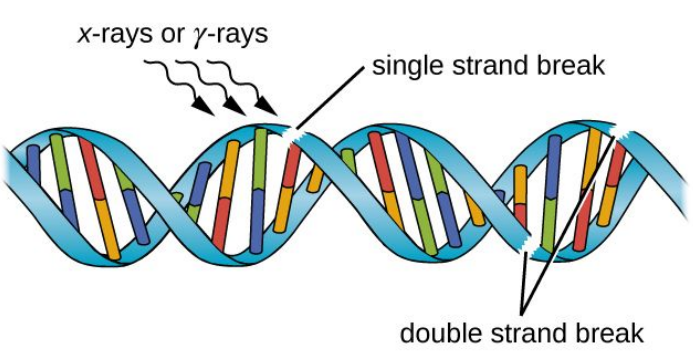
Knock-down/out:
- RNAi
- CRISPR-Cas9
- Morpholinos
- TALENs

Knock-in:
- Transgenesis:
  - Over-expression
  - Ectopic expression
- CRISPR-Cas9 / TALENs

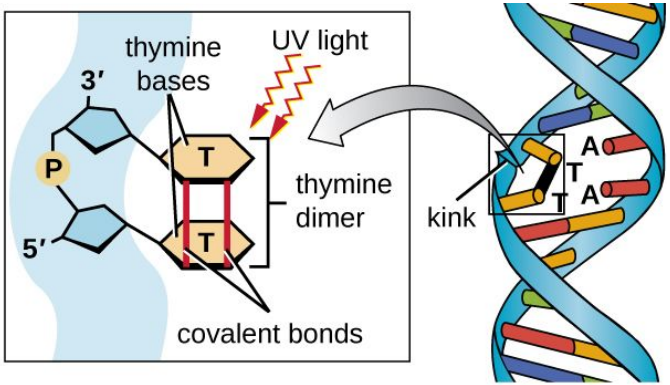
Random mutagenesis (forward genetics) with chemicals, radiation, or mobile elements



Random mutagenesis (forward genetics): ionizing and UV radiation



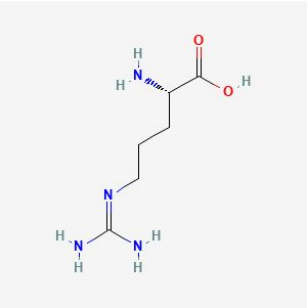
(a) Ionizing radiation



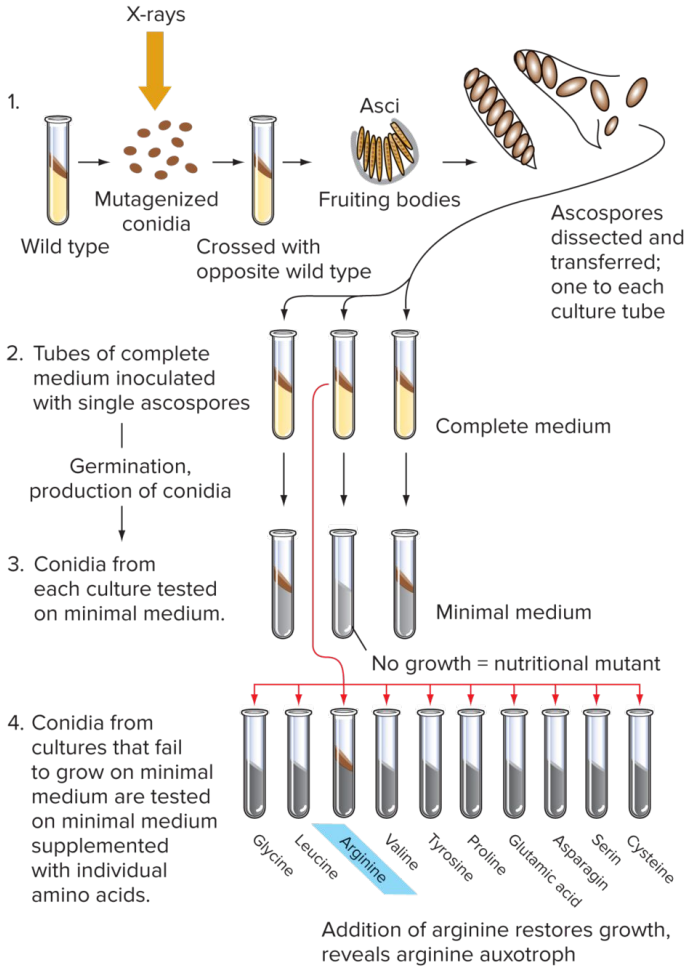
(b) Non-ionizing radiation

Random mutagenesis (forward genetics) revealed one gene encodes one enzyme

Auxotroph = mutant unable to make/utilize a particular nutrient



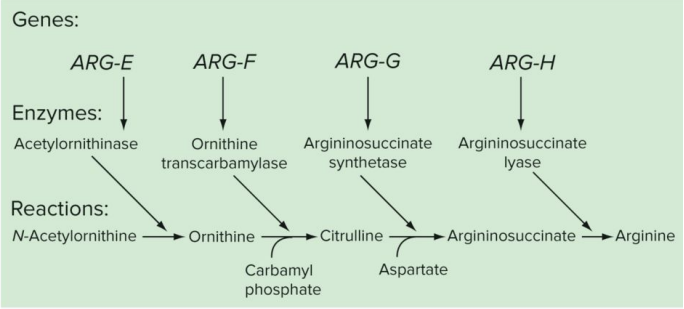
(a) Isolation of arginine auxotrophs



(b) Growth response if nutrient is added to minimal medium

Mutant strain	Supplements: Nothing	Ornithine	Citrulline	Arginino-succinate	Arginine
Wildtype: Arg^+	+	+	+	+	+
ARG-E$^-$	–	+	+	+	+
ARG-F$^-$	–	–	+	+	+
ARG-G$^-$	–	–	–	+	+
ARG-H$^-$	–	–	–	–	+

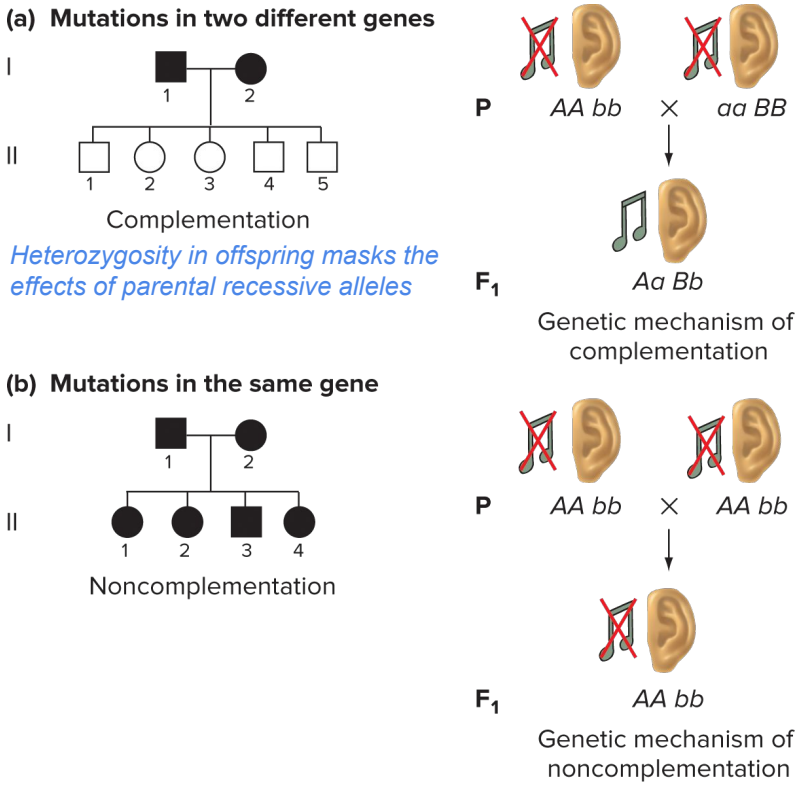
(c) Inferred biochemical pathway



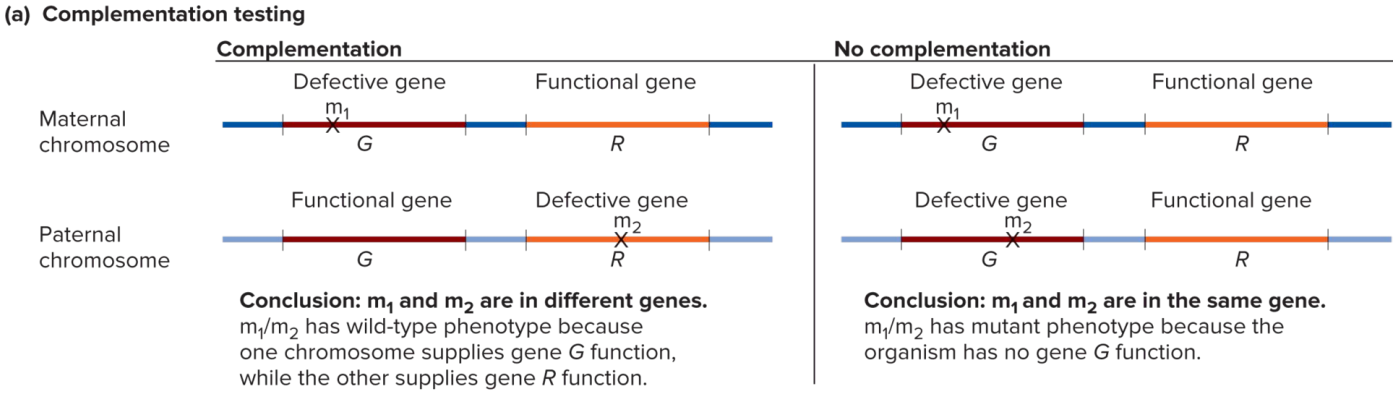
Minimal media-genotype complementation experiments revealed the biochemical pathway

Complementation tests distinguish whether phenotypes are encoded by one gene or more

- Most loss of function mutations are recessive
- Most mutations are rare (<1% allele frequency)
- Therefore, most diploids will be heterozygous for most loss of function mutations, and of wild type phenotype



Complementation tests distinguish whether phenotypes are encoded by more than one gene



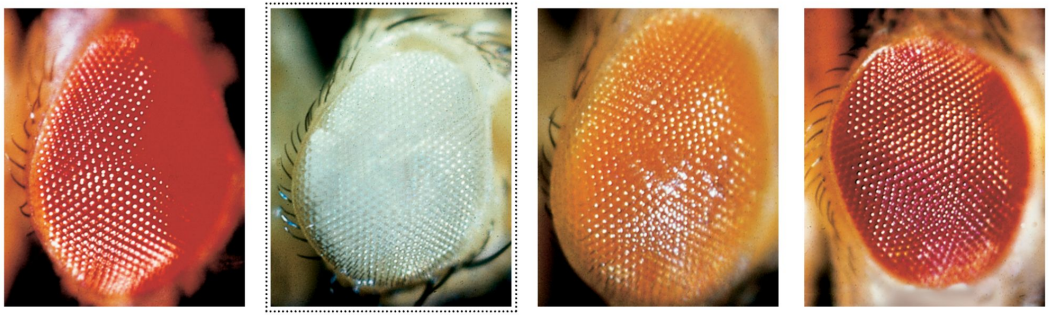
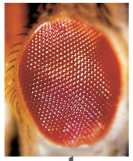


Figure 7.21 *Drosophila* eye color mutations produce a variety of phenotypes. Flies carrying different X-linked eye color mutations. From the *left:* ruby, white, and apricot; a wild-type eye is at the *far right*.

Complementation tests distinguish whether phenotypes are encoded by more than one gene



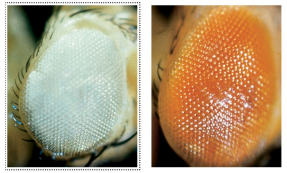
wt

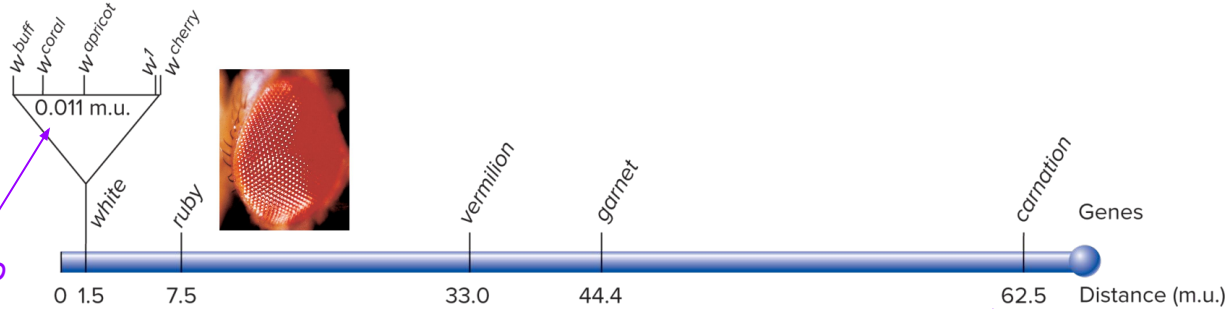
(b) A complementation table: X-linked eye color mutations in *Drosophila*

Alleles of *white*

Mutation	*white*	*garnet*	*ruby*	*vermilion*	*cherry*	*coral*	*apricot*	*buff*	*carnation*
white	–	+	+	+	–	–	–	–	+
garnet		–	+	+	+	+	+	+	+
ruby			–	+	+	+	+	+	+
vermilion				–	+	+	+	+	+
cherry					–	–	–	–	+
coral						–	–	–	+
apricot							–	–	+
buff								–	+
carnation									–

(c) Genetic map: X-linked eye color mutations in *Drosophila*





Alleles at different positions within a gene are less likely to be unlinked by recombination

Recombination broke the association among these alleles

Random mutagenesis (forward genetics): EMS

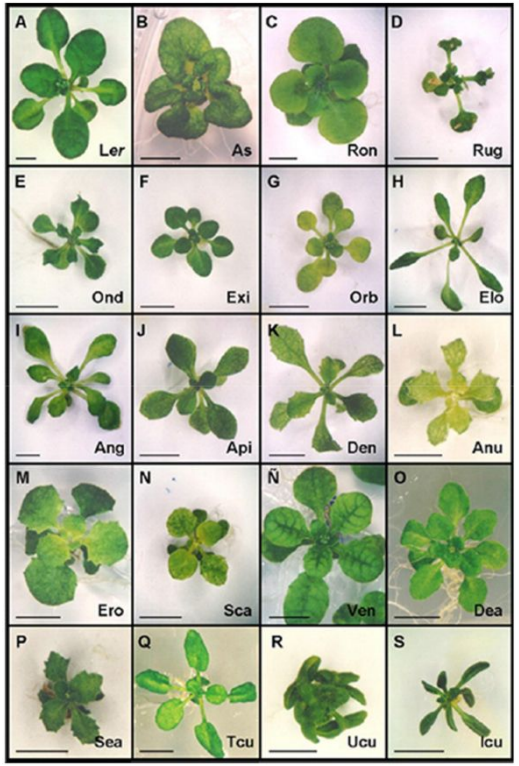
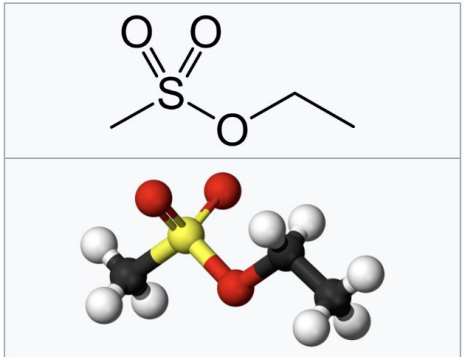
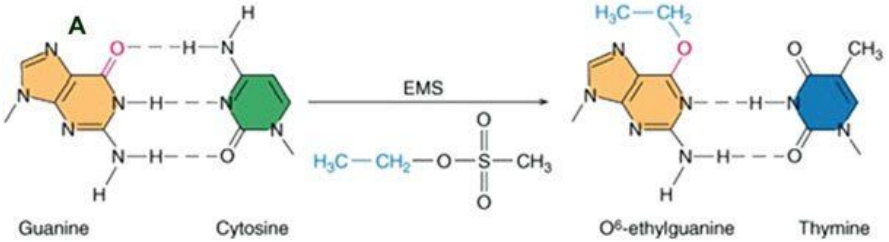


Fig. 5 ***Arabidopsis*** **(L.) Heynh. leaf mutant classes of L*er* ecotype from an EMS-mutagenized collection.** (kindly provided by Prof. J.L. Micol).

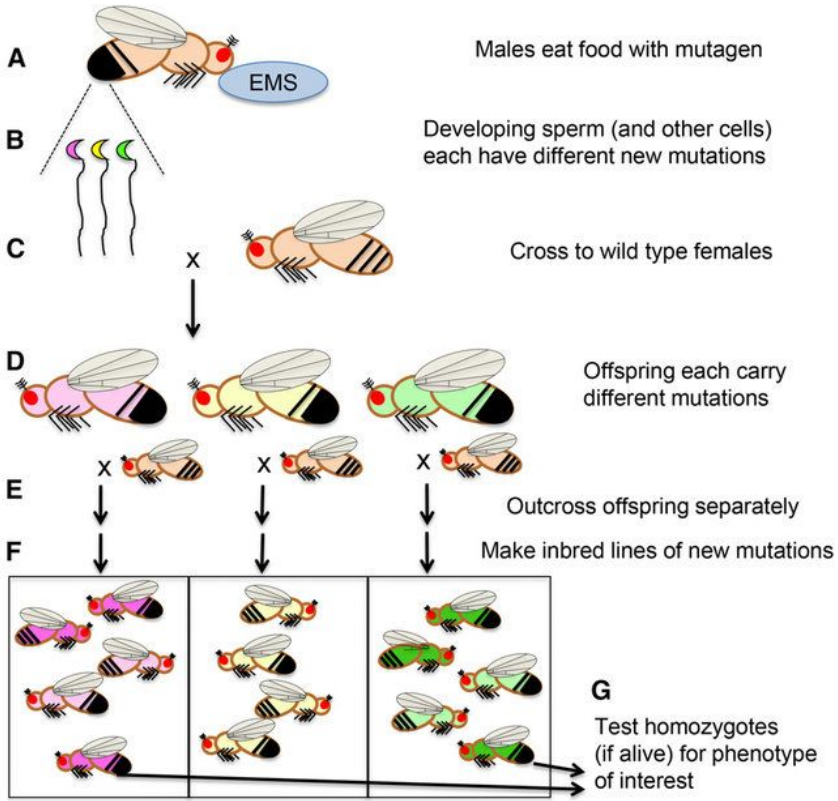
Ethyl methanesulfonate[1]



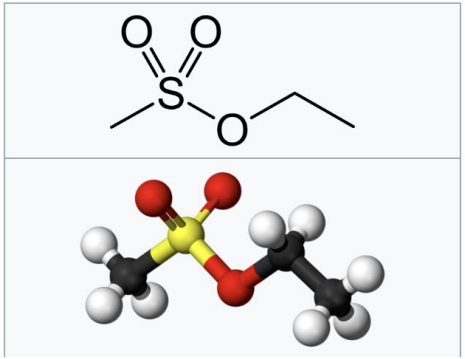


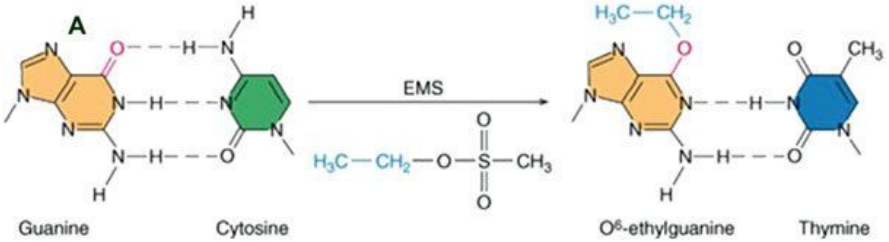
G:C to A:T transitions induced by guanine alkylation

Random mutagenesis (forward genetics): EMS



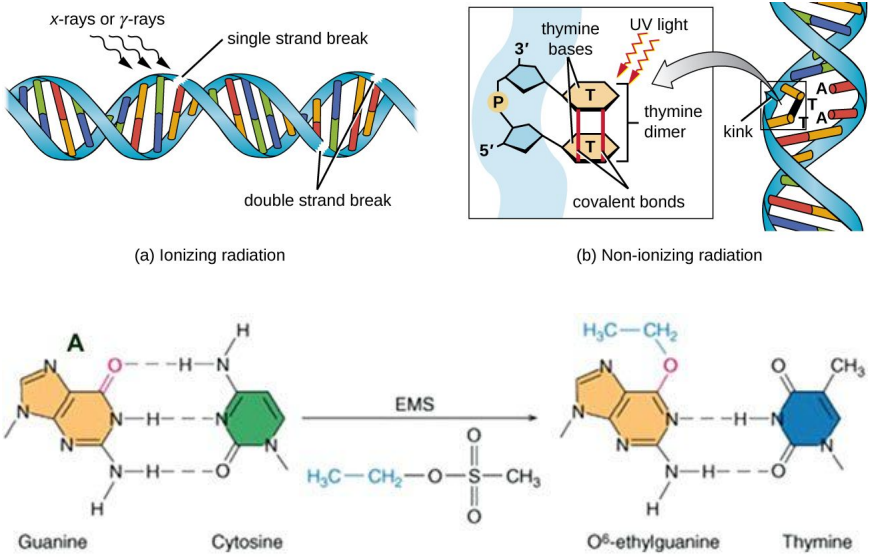
Ethyl methanesulfonate[1]

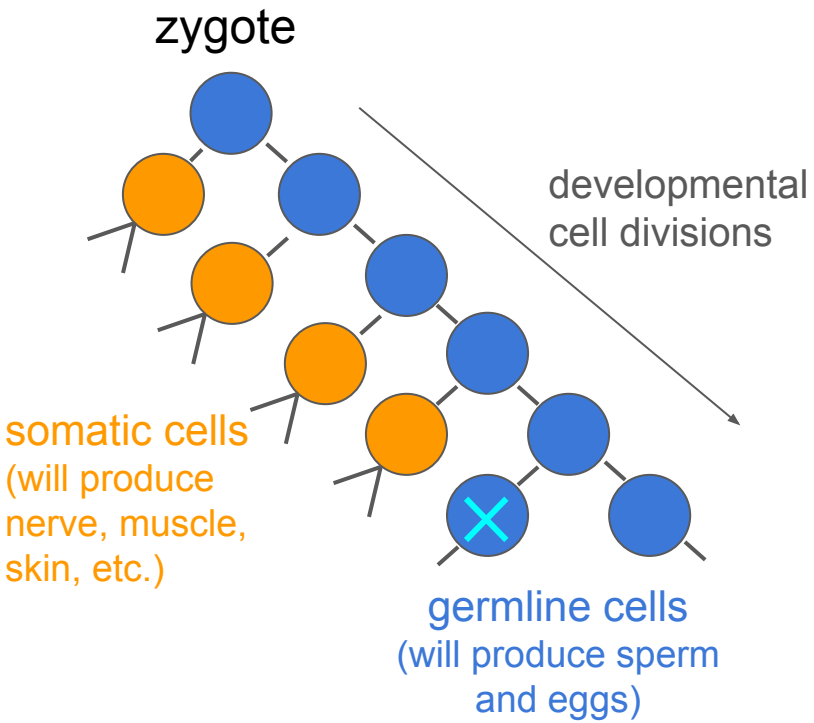




G:C to A:T transitions induced by guanine alkylation

Random mutagenesis (forward genetics): must hit the germline





Transposable elements are called jumping genes

Cut/Copy and paste themselves around the host genome

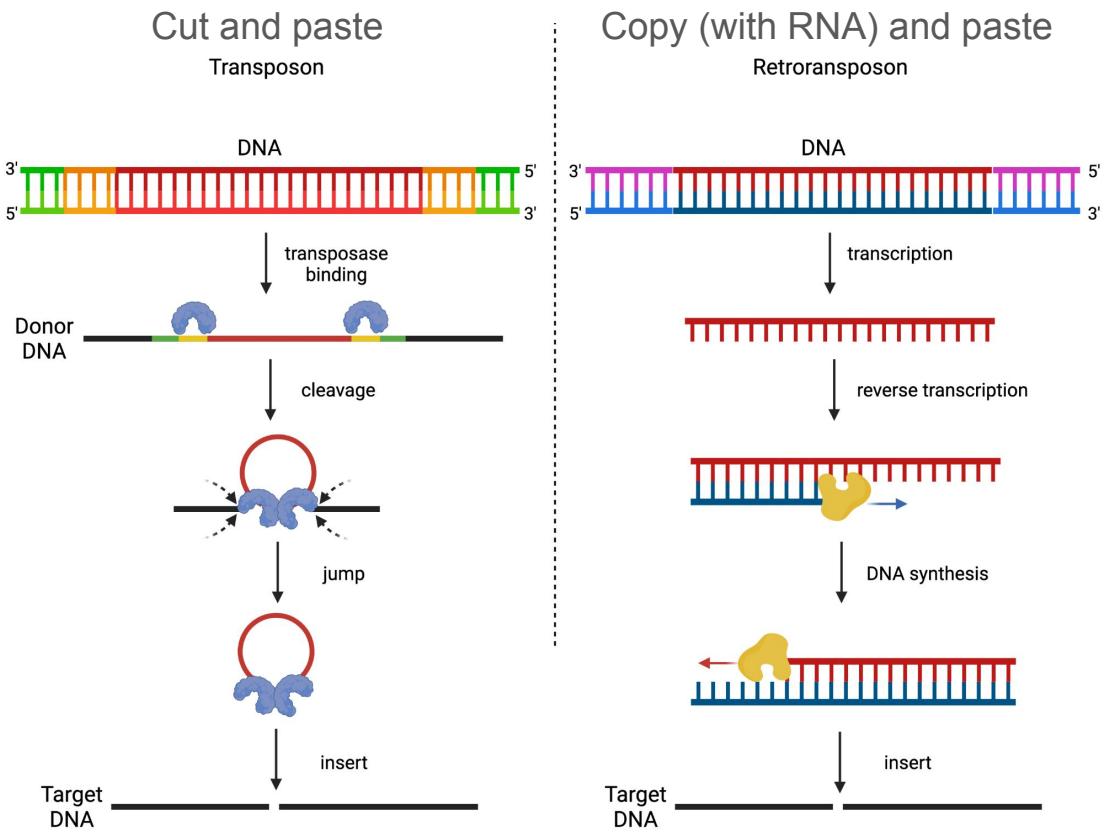
- Autonomously or non-autonomously
- Also called mobile elements

“Selfish” because they encode their own ability to replicate and transmit themselves

Comprise high amounts of eukaryotic genomes (by bp)

Must keep jumping to stay “alive” - mutations erode function

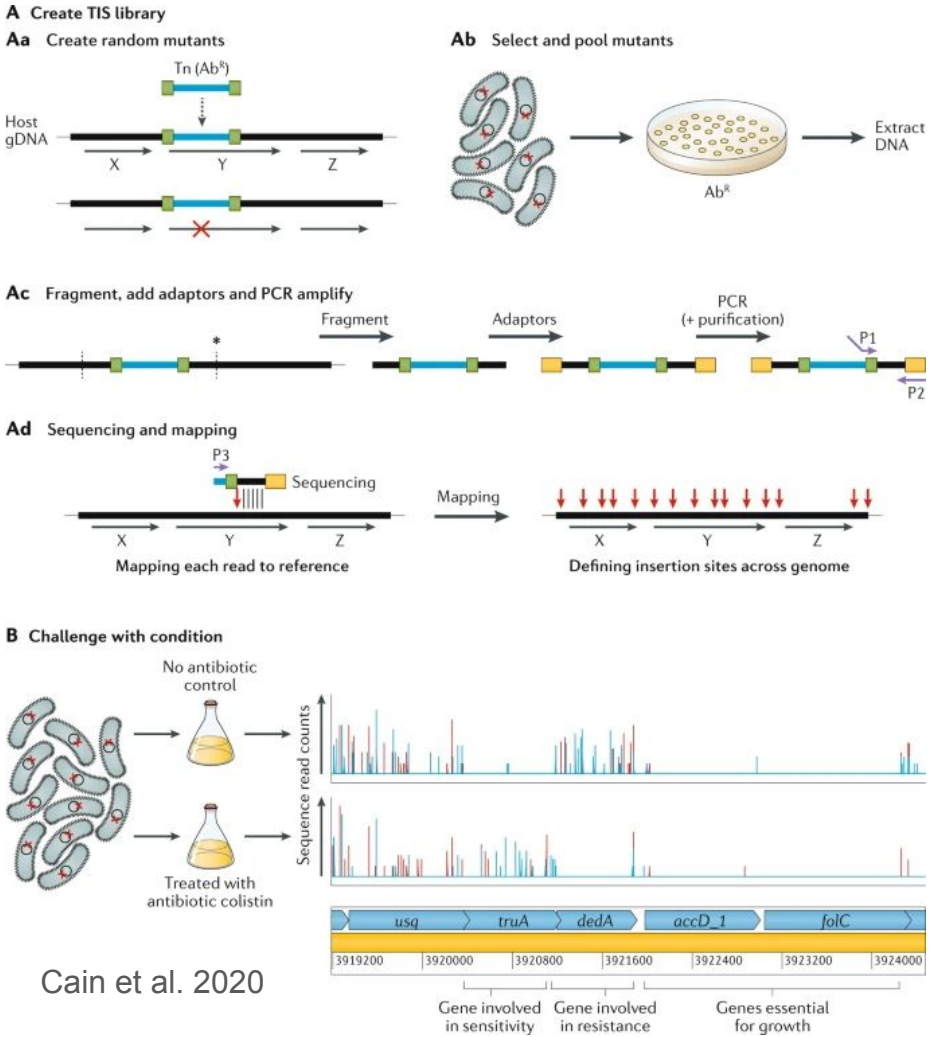
- Host is not incentivized to repair them because jumping is costly



Random mutagenesis (forward genetics): Transposon-insertion mutagenesis

Transposons creates random large insertions genome-wide
- Loss-of function
- Saturation (all genes hit)

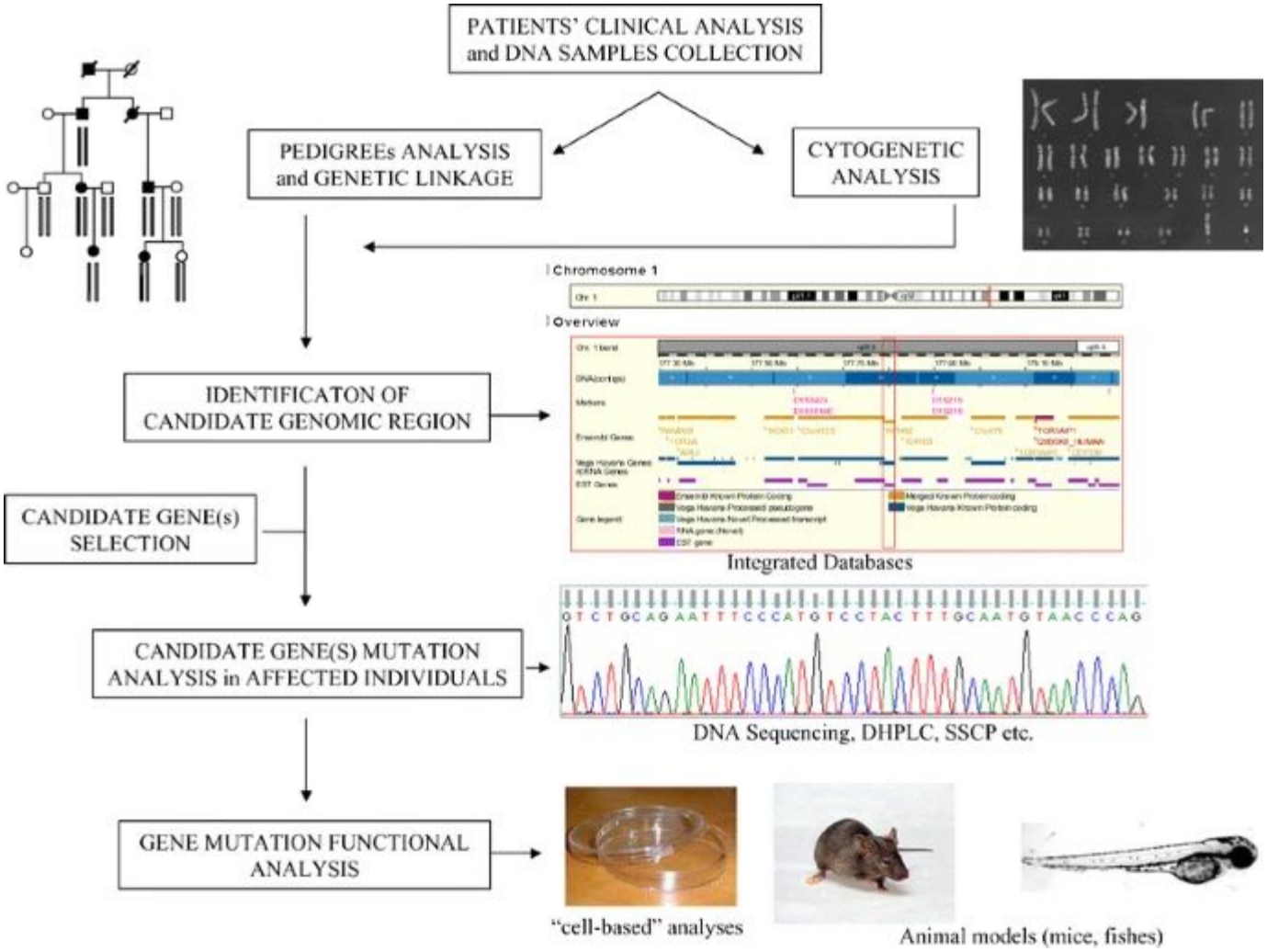
Phenotype via experimentation:
- Assay requires function
- Only mutants retaining function make it through assay



Cain et al. 2020

Overview of positional cloning steps

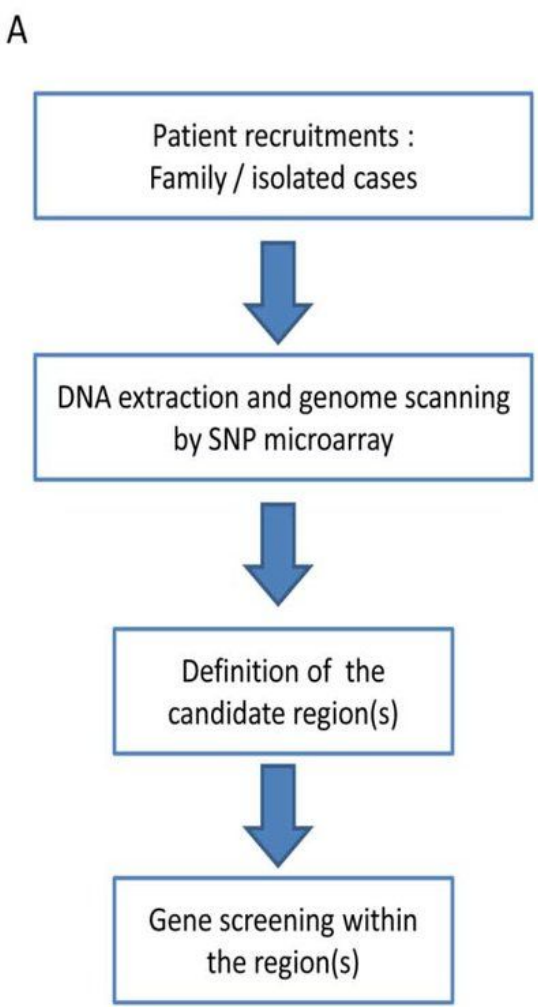
Early forward genetics approach leveraging natural variation

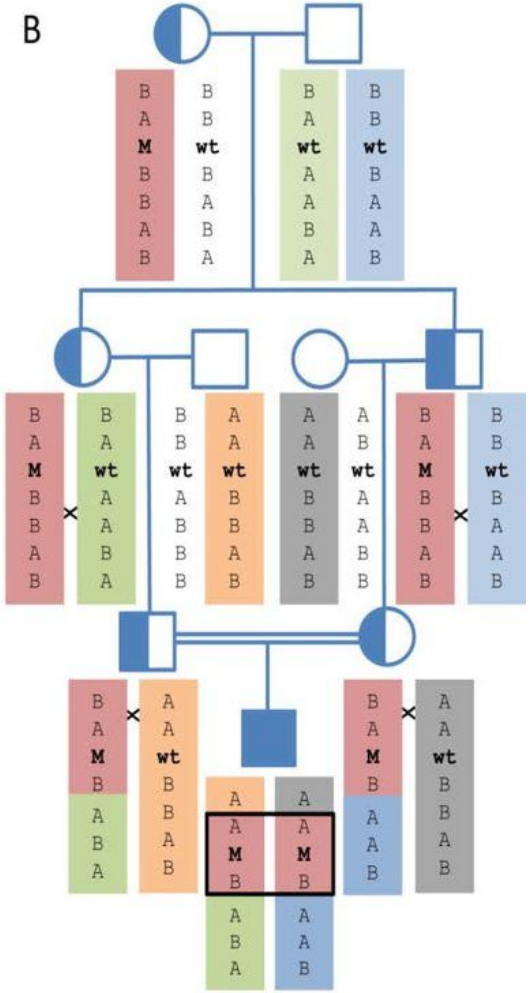


Overview of positional cloning steps

Early forward genetics approach leveraging natural variation

And meiotic recombination

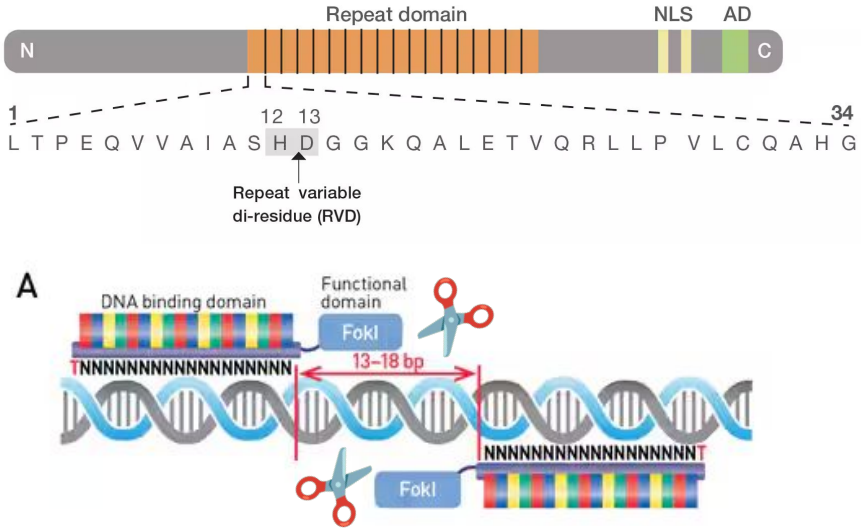




TALEN-based targeted genome editing

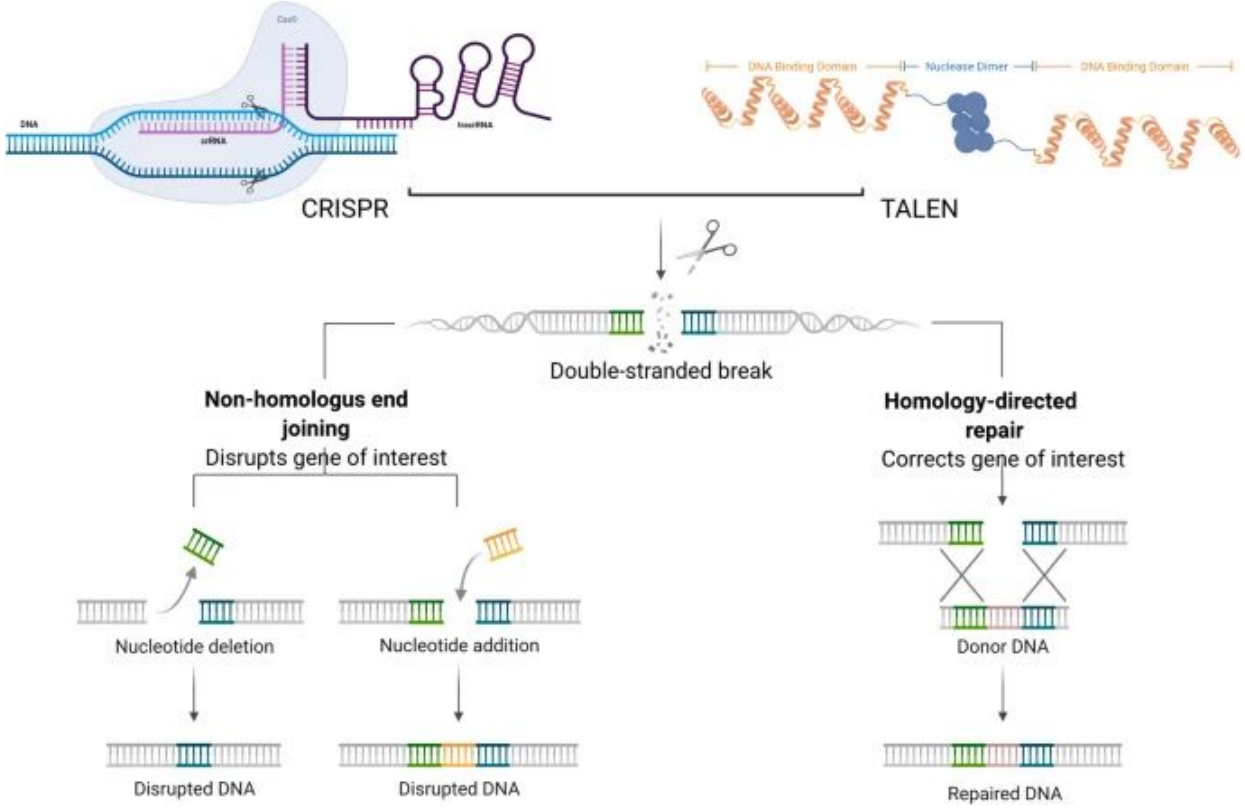
Transcription activator-like effector nucleases (TALEN)

- Restriction enzymes
  - From *Xanthomonas* (bacteria)
- Engineered to cut specific sequences of DNA

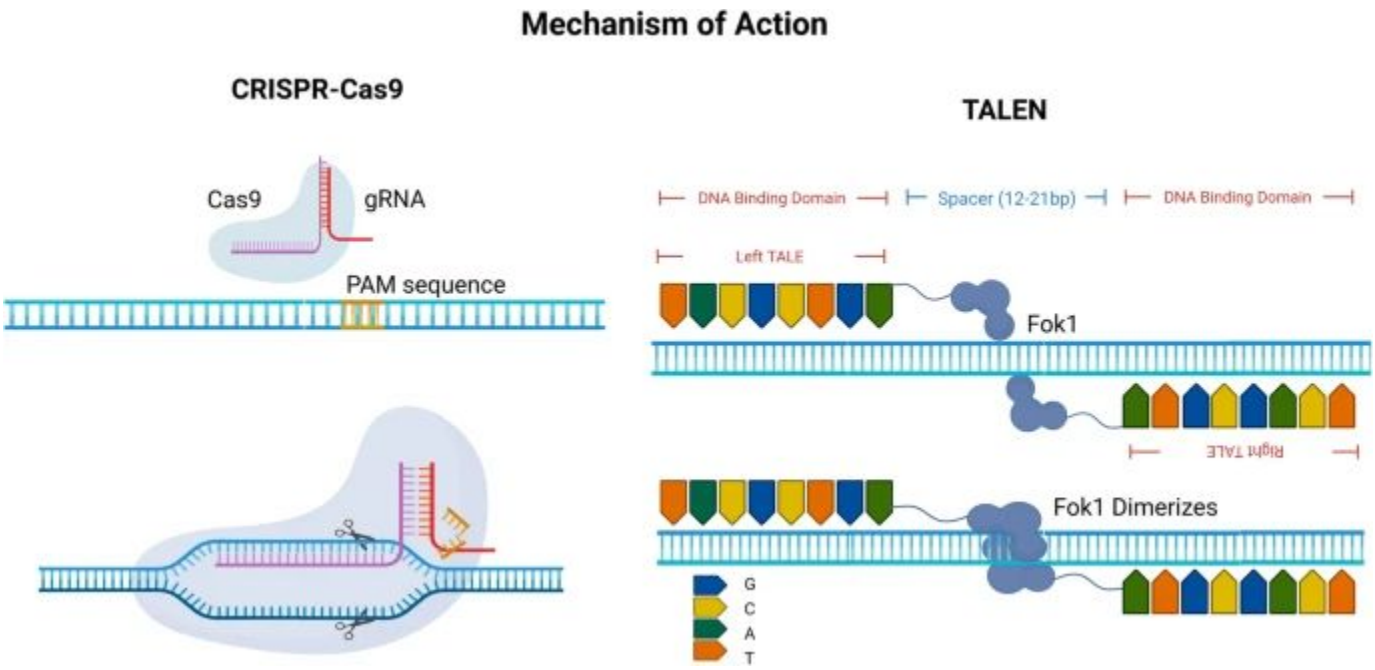


One of the main genome-editing approaches pre-CRISPR (2014)

Downstream of dsDNA cleavage, CRISPR and TALEN are similar (leverage cellular DNA repair mechanisms)

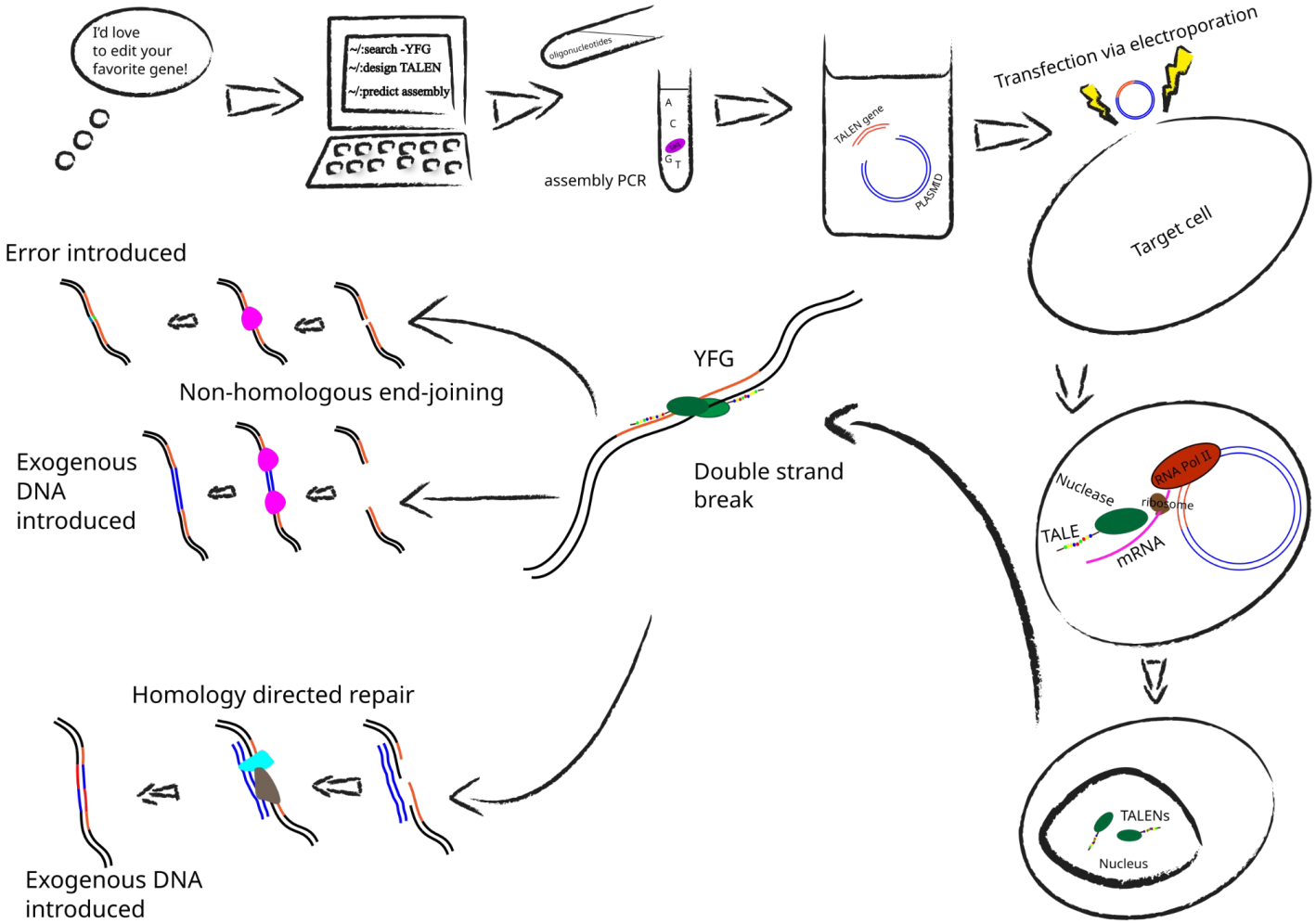


Downstream of dsDNA cleavage, CRISPR and TALEN are similar (leverage cellular DNA repair mechanisms)



Running a TALENs experiment

Transcription Activator-Like Effector Genome Editing



Before the next class:

Problem Set 13

Continue reading in Genetics: From Genes to Genomes

a. Finish Chapter 7.4-7.5
b. Start Chapter 12.1-12.3, Chapter 17.1, 17.2

Come to discussion section and office hours with questions!